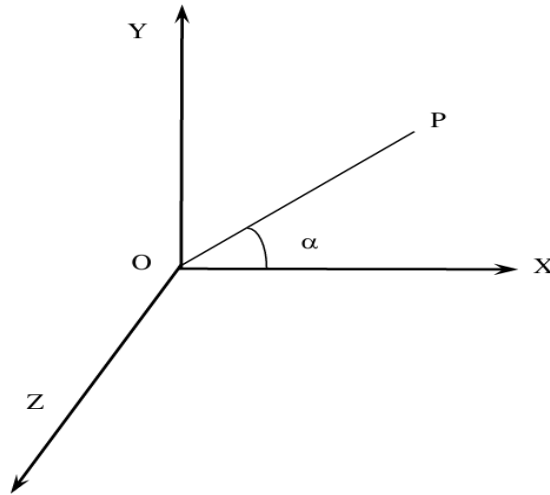


Solid Geometry

Direction Cosine, Direction Ratio

Direction cosines: If a given line OP makes angles α , β , γ with the positive direction of axes of x, y, and z, respectively, then $\cos \alpha$, $\cos \beta$ and $\cos \gamma$ are the direction cosines of the line OP, and are generally denoted by l, m, n respectively.

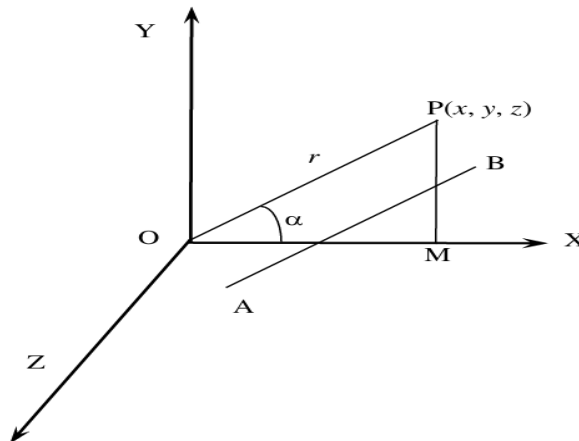
So, $\cos \alpha = l$, $\cos \beta = m$ and $\cos \gamma = n$.



Problem: If l, m, n be the direction cosines of any line, then $l^2 + m^2 + n^2 = 1$,

i.e., the sum of the squares of the direction cosines of every line is one.

Proof: Let OP be drawn through the origin parallel to the given line so that l, m, n are the cosines of angles that make with OX, OY, OZ, respectively.



Let (x, y, z) be the co-ordinates of any point P on this line.

Let $OP = r$.

$$\cos \alpha = OM/OP$$

$$\text{or, } \cos \alpha = x/r$$

Therefore, $x = lr, y = mr, z = nr$.

Squaring and adding, we obtain

$$x^2 + y^2 + z^2 = (l^2 + m^2 + n^2)r^2$$

$$\text{or, } r^2 = (l^2 + m^2 + n^2)r^2$$

$$\text{or, } l^2 + m^2 + n^2 = 1$$

Again,

$$\text{or, } \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma = 1$$

$$\text{or, } 1 - \sin^2 \alpha + 1 - \sin^2 \beta + 1 - \sin^2 \gamma = 1$$

$$\text{or, } 3 - (\sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma) = 1$$

$$\therefore \sin^2 \alpha + \sin^2 \beta + \sin^2 \gamma = 2$$

Direction Ratios: Any three numbers a, b and c which are proportional to the direction cosines l, m, n respectively of a given line are called the direction ratios of the given line (rectangular axes). Now from the definition of direction ratios

$$\frac{l}{a} = \frac{m}{b} = \frac{n}{c} = \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{a^2 + b^2 + c^2}} = \pm \frac{1}{\sqrt{a^2 + b^2 + c^2}}$$

$$\text{Therefore, } l = \pm \frac{a}{\sqrt{a^2 + b^2 + c^2}}, \quad m = \pm \frac{b}{\sqrt{a^2 + b^2 + c^2}}, \quad n = \pm \frac{c}{\sqrt{a^2 + b^2 + c^2}}$$

where the same sign either +ve or -ve is to be chosen throughout.

Direction cosine of a line joining two points:

Let, l, m, n be the direction cosines of a line joining two points $A(x_1, y_1, z_1)$ and $B(x_2, y_2, z_2)$, then

$$\frac{x_2 - x_1}{l} = \frac{y_2 - y_1}{m} = \frac{z_2 - z_1}{n}$$

Angle between two lines:

If l_1, m_1, n_1 and l_2, m_2, n_2 be the direction cosines of any two lines AB and CD, and θ be the angle between them, then

$$\cos \theta = l_1 l_2 + m_1 m_2 + n_1 n_2$$

Condition of perpendicularity:

The lines will be perpendicular if $l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$

Condition of parallelism:

The lines will be parallel if $l_1 = l_2, m_1 = m_2, n_1 = n_2$

Ex. 5. Find the direction cosines of the line which is equally inclined to the axes.

If the line makes angles α, β, γ with the axes then

$$\cos \alpha = \cos \beta = \cos \gamma \therefore \alpha = \beta = \gamma \text{ (given)}$$

$$\text{or } \frac{l}{1} = \frac{m}{1} = \frac{n}{1} = \pm \frac{\sqrt{l^2 + m^2 + n^2}}{\sqrt{1 + 1 + 1}} = \pm \frac{1}{\sqrt{3}}$$

$$\therefore d, c, s \text{ are } \left(\pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}}, \pm \frac{1}{\sqrt{3}} \right)$$

Exercise:

(5. If P and Q are $(2, 3, -6)$ and $(3, -4, 5)$ respectively and O be the origin, find the direction cosines of OP, OQ, PQ. [R.U. 1980]

Ans. $2/7, 3/7, -6/7; 3/5\sqrt{2}, -4/5\sqrt{2}, 1/\sqrt{2}; (-1, 7, -11)/\sqrt{171}$

(6. Find the angle between the lines whose direction ratios are $(3, 1, 2)$ and $(2, -2, 4)$. Ans. $\cos^{-1} \sqrt{3}/7$

18. Show that the lines from A (5, 2, -3) to B (6, 1, 4) and from C (-3, -2, -1) to D (-1, -4, 13) are parallel.

19. Show that the lines AB and BC are perpendicular to each other, where A (-11, 8, 4), B (-1, -7, -1), C (9, -2, 4)

20. Find the direction cosines of the line which is perpendicular to the lines with direction cosines proportional to (1, -2, -2), (0, 2, 1).
Ans. $2/3, -1/3, 2/3$.

24. Prove that the angle between two diagonals of a cube is $\cos^{-1} \frac{1}{3}$

30. If P and Q are the points (2, -3, 4) and (-1, 2, 3) respectively find the direction cosines of OP, OQ and PQ where O is the point (0, 0, 0). [D. U. 1979]

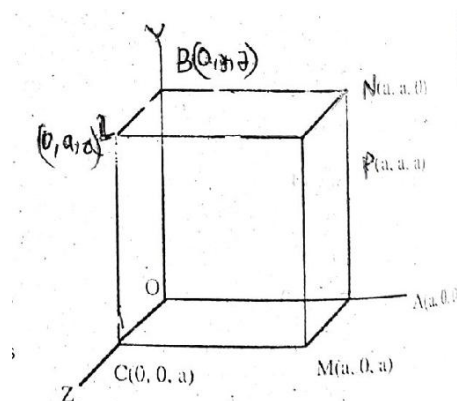
Ans. $(2/\sqrt{29}, -3/\sqrt{29}, 4/\sqrt{29}), (-1/\sqrt{14}, 2/\sqrt{14}, 3/\sqrt{14}), (-3/\sqrt{35}, 5/\sqrt{35}, -1/\sqrt{35})$

31. If the point P and Q are given by (2, 3, 4) and (1, 1, -1) respectively find the angle between OP and OQ. Ans. $1/7\sqrt{3}$. [D. U. 1980]

32. Prove that the joining the points (2, 3, -2) and (3, 1, 1) is parallel to the line joining the points (2, 1, -5) and (4, -3, 1). [D. U. 1980]

Problem: A line makes angles $\alpha, \beta, \gamma, \delta$ with the four diagonals of a cube, prove that

$$\cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}.$$



Solution: Let the length of a cube is a .

Then the vertices of the cube can be taken as $O(0, 0, 0)$, $P(a, a, a)$, $A(a, 0, 0)$, $B(0, a, 0)$, $C(0, 0, a)$, $L(0, a, a)$, $M(a, 0, a)$ and $N(a, a, 0)$.

Therefore, AL, BM, CN and OP are the diagonals of the cube.

The direction cosines of AL are

$$\begin{aligned} & \left(\frac{-a}{\sqrt{(-a)^2 + a^2 + a^2}}, \frac{a}{\sqrt{(-a)^2 + a^2 + a^2}}, \frac{a}{\sqrt{(-a)^2 + a^2 + a^2}} \right) \\ &= \left(\frac{-a}{a\sqrt{3}}, \frac{a}{a\sqrt{3}}, \frac{a}{a\sqrt{3}} \right) \\ &= \left(-\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right) \end{aligned}$$

Similarly, the direction cosines of BM, CN and OP are $\left(\frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$, $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, -\frac{1}{\sqrt{3}} \right)$ and $\left(\frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}}, \frac{1}{\sqrt{3}} \right)$.

Now we assume that the direction cosines of the line be l, m, n .

Thus

$$\cos \alpha = \frac{-l+m+n}{\sqrt{3}}, \cos \beta = \frac{l-m+n}{\sqrt{3}}, \cos \gamma = \frac{l+m-n}{\sqrt{3}} \text{ and } \cos \delta = \frac{l+m+n}{\sqrt{3}}$$

Squaring both sides and then summing up, we get

$$\begin{aligned} & \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta \\ &= \left(\frac{-l+m+n}{\sqrt{3}} \right)^2 + \left(\frac{l-m+n}{\sqrt{3}} \right)^2 + \left(\frac{l+m-n}{\sqrt{3}} \right)^2 + \left(\frac{l+m+n}{\sqrt{3}} \right)^2 \\ &= \frac{1}{3}(-l+m+n)^2 + \frac{1}{3}(l-m+n)^2 + \frac{1}{3}(l+m-n)^2 + \frac{1}{3}(l+m+n)^2 \end{aligned}$$

$$\begin{aligned}
&= \frac{1}{3}(l^2 + m^2 + n^2 - 2lm + 2mn - 2nl) + \frac{1}{3}(l^2 + m^2 + n^2 - 2lm - 2mn + 2nl) \\
&+ \frac{1}{3}(l^2 + m^2 + n^2 + 2lm - 2mn - 2nl) + \frac{1}{3}(l^2 + m^2 + n^2 + 2lm + 2mn + 2nl) \\
&= \frac{4}{3}(l^2 + m^2 + n^2)
\end{aligned}$$

$$= \frac{4}{3} \times 1 \left[\because l^2 + m^2 + n^2 = 1 \right]$$

$$\therefore \cos^2 \alpha + \cos^2 \beta + \cos^2 \gamma + \cos^2 \delta = \frac{4}{3}$$

Problem: Prove that the straight lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $fmn + gnl + hlm = 0$ are **perpendicular** if $\frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$ and **parallel** if $\sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$.

Solution: Given that

$$al + bm + cn = 0 \tag{1}$$

$$fmn + gnl + hlm = 0 \tag{2}$$

From (1), we have

$$\begin{aligned}
cn &= -(al + bm) \\
\therefore n &= -\frac{al + bm}{c}
\end{aligned}$$

Putting this value of n in (2), we get

$$\begin{aligned}
fm \left(-\frac{al + bm}{c} \right) + gl \left(-\frac{al + bm}{c} \right) + hlm &= 0 \\
\Rightarrow -aflm - bfm^2 - agl^2 - bglm + chlm &= 0
\end{aligned}$$

$$\Rightarrow agl^2 + (af + bg - ch)lm + bfm^2 = 0$$

$$\therefore ag\left(\frac{l}{m}\right)^2 + (af + bg - ch)\frac{l}{m} + bf = 0 \quad (3)$$

which is a quadratic equation in $\frac{l}{m}$.

Let $\frac{l_1}{m_1}$ and $\frac{l_2}{m_2}$ be the roots of (3).

Therefore, product of the roots

$$\begin{aligned} \frac{l_1}{m_1} \times \frac{l_2}{m_2} &= \frac{bf}{ag} \\ \Rightarrow \frac{l_1 l_2}{m_1 m_2} &= \frac{f/a}{g/b} \\ \therefore \frac{l_1 l_2}{f/a} &= \frac{m_1 m_2}{g/b} \end{aligned} \quad (4)$$

Similarly,

$$\frac{m_1 m_2}{g/b} = \frac{n_1 n_2}{h/c} \quad (5)$$

From (4) and (5), we can write

$$\frac{l_1 l_2}{f/a} = \frac{m_1 m_2}{g/b} = \frac{n_1 n_2}{h/c} = k \text{ (say)} \quad (6)$$

$$\therefore l_1 l_2 = k \frac{f}{a}, \quad m_1 m_2 = k \frac{g}{b}, \quad n_1 n_2 = k \frac{h}{c}$$

Two lines are perpendicular if

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$\Rightarrow k \left(\frac{f}{a} + \frac{g}{b} + \frac{h}{c} \right) = 0$$

$$\therefore \frac{f}{a} + \frac{g}{b} + \frac{h}{c} = 0$$

Now the lines represented by (3) will be parallel if the roots of (3) are equal.

$$B^2 - 4 \times A \times C = 0$$

$$\therefore (af + bg - ch)^2 - 4 \times ag \times bf = 0$$

$$\Rightarrow (af + bg - ch)^2 = (2\sqrt{af} \sqrt{bg})^2$$

$$\Rightarrow af + bg - ch = \pm 2\sqrt{af} \sqrt{bg}$$

$$\Rightarrow af \pm 2\sqrt{af} \sqrt{bg} + bg = ch$$

$$\Rightarrow (\sqrt{af})^2 \pm 2\sqrt{af} \sqrt{bg} + (\sqrt{bg})^2 = (\sqrt{ch})^2$$

$$\Rightarrow (\sqrt{af} \pm \sqrt{bg})^2 = (\sqrt{ch})^2$$

$$\Rightarrow \sqrt{af} \pm \sqrt{bg} = \pm \sqrt{ch}$$

$$\therefore \sqrt{af} \pm \sqrt{bg} \pm \sqrt{ch} = 0$$

Problem: Prove that the straight lines whose direction cosines are given by the relations $al + bm + cn = 0$ and $ul^2 + vm^2 + wn^2 = 0$ are **perpendicular** if

$a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$ and **parallel** if $\frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$.

Solution: Given that

$$al + bm + cn = 0 \quad (1)$$

$$ul^2 + vm^2 + wn^2 = 0 \quad (2)$$

From (1), we have

$$cn = -(al + bm)$$

$$\therefore n = -\frac{al + bm}{c}$$

Putting this value of n in (2), we get

$$\begin{aligned} ul^2 + vm^2 + w\left(-\frac{al + bm}{c}\right)^2 &= 0 \\ \Rightarrow ul^2 + vm^2 + w\frac{(al + bm)^2}{c^2} &= 0 \\ \Rightarrow c^2ul^2 + c^2vm^2 + w(al + bm)^2 &= 0 \\ \Rightarrow c^2ul^2 + c^2vm^2 + w(a^2l^2 + 2ablm + b^2m^2) &= 0 \\ \Rightarrow (c^2u + a^2w)l^2 + 2abwlm + (c^2v + b^2w)m^2 &= 0 \\ \therefore (c^2u + a^2w)\left(\frac{l}{m}\right)^2 + 2abw\frac{l}{m} + (c^2v + b^2w) &= 0 \end{aligned} \quad (3)$$

which is a quadratic equation in $\frac{l}{m}$.

Let $\frac{l_1}{m_1}$ and $\frac{l_2}{m_2}$ be the roots of (3).

Therefore, product of the roots

$$\frac{l_1}{m_1} \times \frac{l_2}{m_2} = \frac{c^2v + b^2w}{c^2u + a^2w}$$

$$\Rightarrow \frac{l_1 l_2}{m_1 m_2} = \frac{c^2 v + b^2 w}{c^2 u + a^2 w}$$

$$\therefore \frac{l_1 l_2}{c^2 v + b^2 w} = \frac{m_1 m_2}{c^2 u + a^2 w} \quad (4)$$

Similarly,

$$\frac{m_1 m_2}{c^2 u + a^2 w} = \frac{n_1 n_2}{b^2 u + a^2 v} \quad (5)$$

From (4) and (5), we can write

$$\frac{l_1 l_2}{c^2 v + b^2 w} = \frac{m_1 m_2}{c^2 u + a^2 w} = \frac{n_1 n_2}{b^2 u + a^2 v} = k \text{ (say)} \quad (6)$$

$$\therefore l_1 l_2 = k(c^2 v + b^2 w), \quad m_1 m_2 = k(c^2 u + a^2 w), \quad n_1 n_2 = k(b^2 u + a^2 v)$$

Two lines are perpendicular if

$$l_1 l_2 + m_1 m_2 + n_1 n_2 = 0$$

$$\Rightarrow k(c^2 v + b^2 w) + k(c^2 u + a^2 w) + k(b^2 u + a^2 v) = 0$$

$$\Rightarrow k(c^2 v + b^2 w + c^2 u + a^2 w + b^2 u + a^2 v) = 0$$

$$\Rightarrow c^2 v + b^2 w + c^2 u + a^2 w + b^2 u + a^2 v = 0$$

$$\therefore a^2(v + w) + b^2(w + u) + c^2(u + v) = 0$$

Now the lines represented by (3) will be parallel if the roots of (3) are equal.

$$\therefore (2abw)^2 - 4 \times (c^2 u + a^2 w) \times (c^2 v + b^2 w) = 0$$

$$\Rightarrow 4a^2b^2w^2 - 4(c^2u + a^2w)(c^2v + b^2w) = 0$$

$$\Rightarrow a^2b^2w^2 - (c^4uw + b^2c^2uw + a^2c^2vw + a^2b^2w^2) = 0$$

$$\Rightarrow a^2b^2w^2 - c^4uw - b^2c^2uw - a^2c^2vw - a^2b^2w^2 = 0$$

$$\Rightarrow c^4uw + b^2c^2uw + a^2c^2vw = 0$$

$$\Rightarrow a^2vw + b^2uw + c^2uv = 0$$

$$\therefore \frac{a^2}{u} + \frac{b^2}{v} + \frac{c^2}{w} = 0$$
